



A Robust Multi Criteria Framework for Assessing Agricultural Technology Competitiveness and Sustainability: Evidence from East Java, Indonesia

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Abstract: This study aims to evaluate and rank regional agricultural technology competitiveness in East Java, Indonesia, using a structured multi-criteria decision-making approach. Specifically, it addresses four key objectives: (1) to apply the Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) method to assess and rank regional competitiveness across multiple technological dimensions; (2) to examine whether agricultural technology adoption levels differ significantly across regions using one-way Analysis of Variance (ANOVA); (3) to evaluate the sensitivity and robustness of the ranking results under alternative weighting scenarios through sensitivity analysis and rank correlation measures (Spearman's ρ and Kendall's τ); and (4) to derive policy-relevant and system-oriented implications for enhancing competitiveness and reducing regional disparities. The study employs a quantitative approach based on primary survey data collected from 210 farmers across seven regions in East Java. Four key dimensions are considered, namely environmental, irrigation, marketing, and production technologies. The PROMETHEE method is used to generate regional rankings, while ANOVA is applied to test for statistically significant differences in technology adoption. Robustness is further assessed through systematic weight variations and rank correlation analysis. The results reveal substantial regional disparities in relative technological competitiveness, with leading regions demonstrating more balanced, integrated adoption across multiple technological dimensions. ANOVA results confirm that differences in technology adoption across regions are statistically significant ($p < 0.01$), thereby providing complementary statistical evidence for inter-regional variation in the underlying technology adoption indicators used in the PROMETHEE analysis. The robustness analysis shows that the ranking results are highly stable across most weighting scenarios, with only minor variations observed when marketing-related criteria are emphasized. This study contributes methodologically by integrating multi-criteria decision-making with statistical validation and robustness testing in a unified framework. From a policy perspective, the findings highlight the importance of strengthening market access, improving technological integration, and implementing region-specific interventions to enhance agricultural competitiveness and reduce disparities.

Keywords: Agricultural technology competitiveness; Preference Ranking Organization Method for Enrichment Evaluation method; Multi-criteria decision-making; Robustness analysis; Regional disparities

1. Introduction

Agriculture remains a fundamental pillar of economic and social stability, particularly in developing countries. However, its sustainability performance continues to face significant challenges. At the global level, the sector is under increasing pressure from climate change, land degradation, population growth, and rising food demand, all of which threaten the resilience of food systems (FAO, 2022; IPCC, 2022). Recent studies further emphasize that

climate variability, resource depletion, and supply chain disruptions are intensifying risks to global food security and agricultural sustainability (Ortiz-Bobea et al., 2021; Springmann et al., 2023; Zeng et al., 2017). It is widely recognized that food production must increase substantially by 2050 while simultaneously reducing environmental impacts and enhancing adaptive capacity (Searchinger et al., 2019). These challenges highlight the urgency of improving agricultural sustainability through more effective technological and institutional interventions.

In Indonesia, agriculture plays a vital role in employment and rural economic development. Despite this importance, recent indicators reveal structural weaknesses in sustainability performance, particularly in water management and climate adaptation. Similar patterns have been observed across emerging economies, where institutional capacity and governance challenges limit the effectiveness of climate adaptation strategies (Barrett, 2021; OECD/FAO, 2021). These limitations reflect systemic constraints that hinder the agricultural sector's ability to respond effectively to environmental and economic changes.

Digital and modern agricultural technologies are increasingly recognized as key drivers of productivity, food security, and sustainable development (Feng et al., 2022; Klerkx & Rose, 2020). Recent literature highlights the growing role of digital agriculture, artificial intelligence, and precision farming in enhancing decision-making and resource efficiency (Li et al., 2023; Wolfert et al., 2023). However, the adoption of these technologies among smallholder farmers remains uneven. Constraints such as limited access to technology, low digital literacy, financial barriers, and uneven institutional support continue to impede widespread adoption (Aker et al., 2023; Tesfaye et al., 2021). This gap between technological potential and actual adoption remains a key challenge in achieving sustainable agricultural transformation.

Existing studies on agricultural technology adoption have identified various determinants, including farmer characteristics, access to resources, and institutional factors (Alwang et al., 2017; Mwangi & Kariuki, 2015). While these studies provide valuable insights, they often focus on individual technologies or isolated adoption decisions. More recent research has begun to examine the joint adoption of multiple technologies and their interactions; however, findings remain inconsistent, particularly regarding how combinations of technologies influence productivity and sustainability outcomes (Amadu et al., 2020; Tufa et al., 2019). This indicates a need for a more comprehensive and structured approach to evaluating agricultural technology adoption.

In the Indonesian context, technology adoption is influenced by multiple interacting factors, including technology attributes, farm management practices, and broader socio-economic conditions (Bennett et al., 2013; Sehnem et al., 2019). Although various stakeholders, including government agencies, research institutions, and private actors, continue to promote agricultural innovations, many smallholder farmers still rely on traditional practices and have limited access to modern technologies (Braun & Ghosh, 2020). Recent evidence also suggests that the adoption rate of agricultural technologies has not yet reached national targets, indicating a persistent gap between policy ambitions and field-level implementation. Similar challenges are observed in other developing countries, where adoption remains uneven despite increasing availability of innovations (Gamage et al., 2023; Pham et al., 2021).

In East Java, these challenges are reflected in regional disparities in agricultural performance. Differences in productivity, access to resources, and technology adoption suggest the presence of structural inequalities that require closer examination. While previous studies have explored aspects of technology use and farmer behavior, they tend to rely on descriptive approaches and do not systematically evaluate the relative importance or prioritization of different technologies. Recent literature emphasizes the importance of multi-criteria decision-making frameworks for addressing complex agricultural challenges that involve trade-offs among economic, environmental, and social objectives (Behzadian et al., 2010; Cinelli et al., 2014; Kumar & Pant, 2022; Zhou et al., 2020). However, their application in regional agricultural technology prioritization remains limited.

This study addresses these gaps by proposing a multi-criteria evaluation framework to assess agricultural technology competitiveness across regions. The study applies the Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) method to evaluate multiple technology dimensions in an integrated manner (Behzadian et al., 2010; Brans & Vincke, 1985). Specifically, it contributes to the literature in three key ways. First, methodologically, it integrates the PROMETHEE approach with statistical validation (ANOVA) and sensitivity analysis, providing a more robust and reliable assessment framework. Second, empirically, it offers a comparative evaluation of technological competitiveness across regions in East Java, capturing spatial heterogeneity in multi-dimensional technology adoption. Third, conceptually, it extends conventional multi-criteria analysis by incorporating a feedback-loop perspective on the relationships among technology adoption, productivity, and sustainability outcomes. However, the PROMETHEE results in this study primarily represent relative regional technological competitiveness based on self-reported technology adoption levels rather than directly measured productivity, income, environmental performance, or sustainability outcomes. By combining quantitative rigor with conceptual interpretation, this study moves beyond isolated technology assessments and provides a more comprehensive understanding of how multiple technological dimensions jointly shape regional agricultural competitiveness. Accordingly, this study addresses the following research questions:

1. How can the PROMETHEE method be applied to evaluate and rank regional agricultural technological competitiveness in East Java, Indonesia, based on multiple technology-adoption criteria?

2. To what extent do agricultural technology adoption levels differ significantly across regions, as validated using one-way ANOVA?
3. How sensitive and robust are the regional competitiveness rankings to variations in criteria weights, as assessed through sensitivity analysis and rank correlation measures (Spearman's ρ and Kendall's τ)?
4. What policy-relevant and system-oriented implications can be derived to enhance agricultural technology competitiveness and reduce regional disparities?

2. Literature Review and Research Framework

2.1 Agricultural Technology and Competitiveness

Agricultural technology plays a central role in enhancing productivity, efficiency, and sustainability in farming systems. The adoption of modern technologies, including improved seed varieties, irrigation systems, and digital tools, has been widely recognized as a key driver of agricultural competitiveness (Feng et al., 2022; Mwangi & Kariuki, 2015). Recent studies further emphasize that technological innovation contributes not only to increased yields but also to improved resource efficiency and environmental sustainability (Gamage et al., 2023; Li et al., 2023). Competitiveness in agriculture is increasingly understood as a multidimensional concept that encompasses productivity, environmental performance, and market integration. This perspective aligns with the framework used in this study, which evaluates technology adoption across four dimensions: environmental, irrigation, marketing, and production technologies. These dimensions reflect the integrated nature of agricultural systems, where improvements in one domain may influence outcomes in others.

2.2 Determinants of Agricultural Technology Adoption

The literature identifies multiple factors influencing farmers' decisions to adopt agricultural technologies. These include socio-economic characteristics (e.g., age, education, farm size), access to financial resources, institutional support, and market conditions (Alwang et al., 2017; Mwangi & Kariuki, 2015). In developing countries, smallholder farmers often face structural constraints such as limited access to credit, low levels of technical knowledge, and inadequate extension services, which restrict their ability to adopt new technologies (Aker et al., 2023; Tesfaye et al., 2021). Recent studies also highlight the role of digital literacy and infrastructure in shaping technology adoption, particularly in the context of digital agriculture (Klerkx & Rose, 2020; Wolfert et al., 2023). These findings support the conceptual framework of this study, which positions farmers as central actors whose adoption decisions are shaped by contextual conditions.

2.3 Multi-Technology Adoption and Interaction Effects

While early studies focused on single-technology adoption, more recent research has examined the adoption of multiple technologies and their interaction effects. Approaches such as joint estimation models and technology bundling have been used to analyze how different technologies are adopted simultaneously and how they influence agricultural outcomes (Amadu et al., 2020; Tufa et al., 2019). However, the findings remain mixed. Some studies suggest that combining technologies leads to synergistic effects, while others indicate diminishing returns or context-specific outcomes (Hörner et al., 2022). These inconsistencies highlight the need for analytical frameworks that can systematically evaluate multiple technologies and their relative importance, an issue directly addressed in this study through a multi-criteria approach.

3. Methodology

3.1 Data Collection and Sampling Design

This study employs a structured multi-stage sampling design that integrates cluster, stratified, and purposive techniques to ensure analytical relevance and comparability across regions. First, cluster sampling was applied to classify regions into three categories: high, medium, and low levels of agricultural technology adoption based on secondary data from regional agricultural statistics. This classification provides a theoretically informed basis for capturing heterogeneity in technological development and ensures that the analysis reflects variation in regional competitiveness. Second, stratified sampling was implemented within each selected region to improve internal representativeness. Farmers were grouped based on key characteristics, including farming experience, land size, and participation in agricultural programs. This step minimizes within-region sampling bias and ensures that different farmer profiles are adequately represented in the dataset. Third, purposive sampling was used to select respondents who met predefined inclusion criteria: (1) actively engaged in farming activities, (2) having prior exposure to agricultural technologies, and (3) willingness to participate in the survey. This approach ensures that

respondents possess sufficient knowledge and experience relevant to the study objectives, thereby improving data validity.

A fixed quota of 30 households per region was applied, resulting in a total sample of 210 respondents. This equal-allocation strategy prioritizes cross-regional comparability over proportional representation. Consequently, the sampling design is best characterized as a quota-based purposive sampling framework embedded within a stratified cluster structure. While this approach limits statistical generalization to the broader population, it is methodologically appropriate for comparative multi-criteria analysis, where consistency across units is essential. The sampling frame was constructed from farmer group registries obtained from local agricultural offices, ensuring that only active, registered farmers were included. This enhances data reliability and reduces the likelihood of including non-representative respondents.

From an analytical perspective, the use of one-way ANOVA in this study is intended to identify statistically significant differences in technology adoption across regions rather than to establish causal inference. Given the non-probability sampling design, the statistical results should be interpreted as indicative of relative inter-regional disparities rather than population-level generalizations. This approach is consistent with prior studies that emphasize the use of inferential statistics in comparative frameworks with controlled sampling structures. To further enhance methodological rigor, the study integrates multi-criteria decision-making (PROMETHEE) II with statistical validation (ANOVA) and robustness testing (sensitivity analysis and rank correlation). This triangulation approach strengthens the internal consistency of the findings and reduces the risk of method-specific bias.

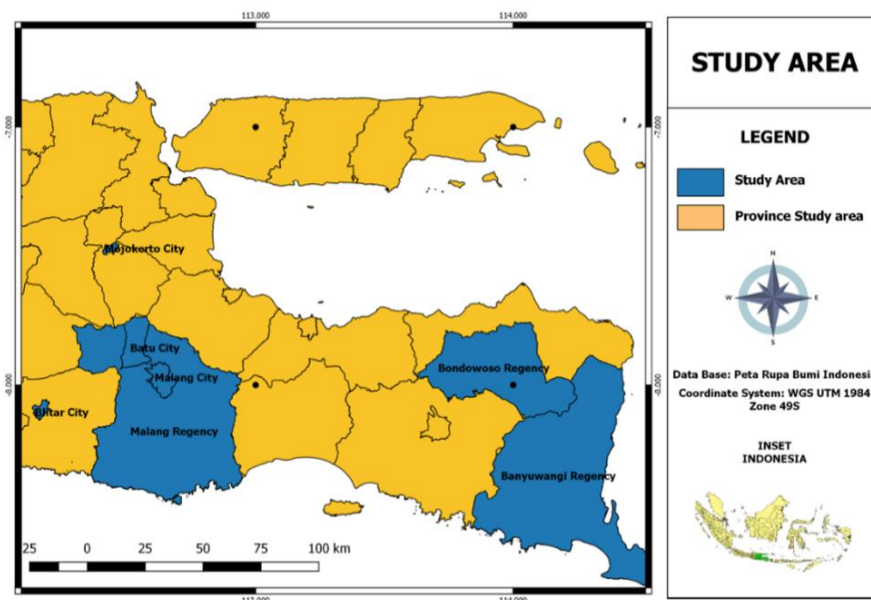


Figure 1. Geographical location and jurisdiction of the case-study area of East Java Province

Figure 1 illustrates the spatial distribution and administrative boundaries of the selected regions in East Java Province. The map provides a geographical justification for the sampling design by showing that the selected regions are distributed across different spatial clusters, representing varying levels of agricultural technology adoption. Importantly, the figure highlights that some regions with contrasting competitiveness levels, such as Batu City and Malang Regency, are geographically proximate. This spatial contrast strengthens the analytical argument that differences in technological competitiveness may not solely reflect geographic location but are more likely influenced by structural factors such as institutional capacity, access to technology, and market integration. Thus, Figure 1 supports both the validity of the cluster-based sampling approach and the interpretation of inter-regional disparities, linking spatial context with empirical findings.

3.2 Analytical Framework

This study adopts an integrated analytical framework in which the PROMETHEE method serves as the core tool for evaluating and ranking regional agricultural technology competitiveness based on multiple criteria. Figure 2 presents the conceptual framework for interpreting the relationships among technology adoption, productivity, and sustainability outcomes. The framework is derived from theoretical considerations and supported by empirical patterns observed in the results, rather than direct causal estimation. Accordingly, the analytical process follows a sequential structure.

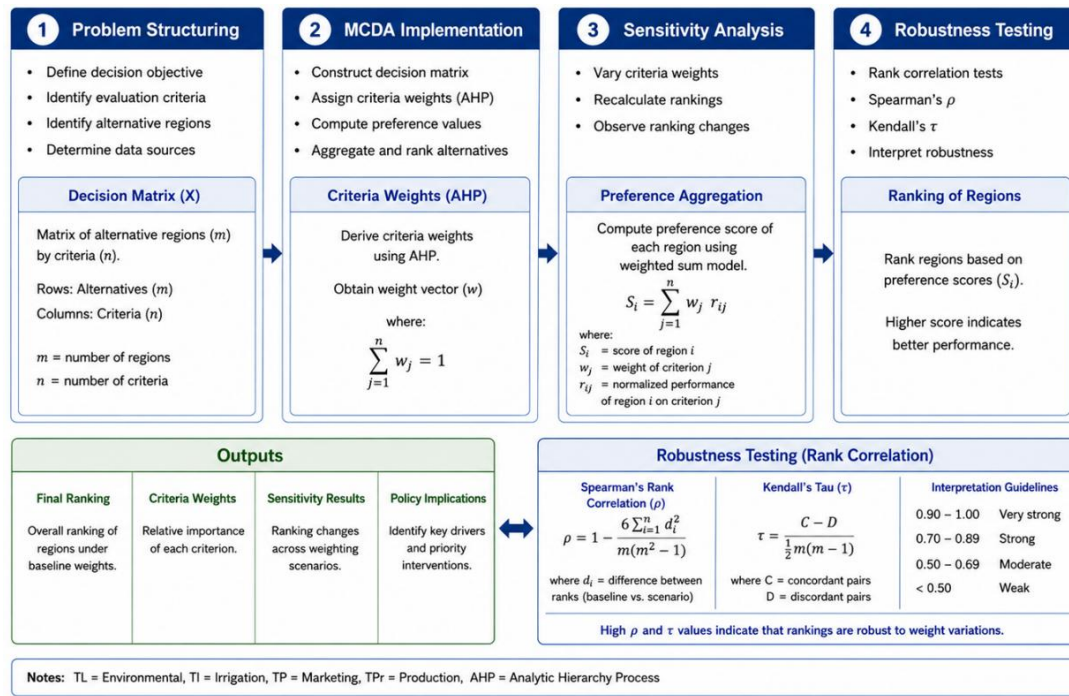


Figure 2. Methodological for multi-criteria decision analysis and robustness evaluation

It is important to emphasize that the PROMETHEE net flow values generated in this study do not directly measure agricultural productivity, farmer income, environmental outcomes, or sustainability performance. Instead, the PROMETHEE outputs represent relative regional technological competitiveness derived from composite scores of self-reported technology adoption indicators across four technological dimensions. Therefore, the ranking results should be interpreted as comparative measures of technology adoption intensity and integration rather than direct measures of sustainability achievement.

First, the PROMETHEE method is applied to evaluate and rank regional technological competitiveness. This approach enables pairwise comparisons across all criteria between regions. In this study, a usual preference function is employed, equal weights are assigned to all criteria, and net flow values (Φ) are used to determine the final ranking. Second, statistical validation is conducted using one-way ANOVA to examine whether differences in technology adoption across regions are statistically significant. The ANOVA model tests for differences in mean values across regions at the 5% significance level. Third, sensitivity analysis is performed to assess the stability of the PROMETHEE ranking results under alternative weighting scenarios. Criteria weights are varied within a defined range ($\pm 10\%$ and $\pm 20\%$), and the resulting changes in rankings are analyzed to identify the most influential criteria and the degree of ranking stability. Fourth, robustness testing is conducted using rank correlation measures, namely Spearman's ρ and Kendall's τ , to evaluate the consistency of rankings across different scenarios. This step ensures that the results are not sensitive to moderate changes in model assumptions, thereby strengthening the reliability of the analytical framework.

3.3 Measurement of Variables

Agricultural technology adoption is measured using four key dimensions: Environmental Technology (TL), Irrigation Technology (TI), Marketing Technology (TP), and Production Technology (TPr). Each dimension is operationalized through a set of indicators capturing different aspects of technology use, including adoption intensity, technical capability, and market-oriented practices. Data were collected using a five-point Likert scale (1 = very low adoption, 5 = very high adoption), enabling standardized comparisons across respondents and regions. The use of a Likert scale is appropriate for capturing perception-based and behavioral aspects of technology adoption in agricultural settings. To ensure analytical consistency, indicator scores within each dimension were aggregated to the mean to construct composite indices. While this approach may reduce individual-level variability, it facilitates comparability across regions and supports multi-criteria analysis. Although this study focuses on comparative analysis rather than construct validation, the selected indicators are derived from established literature on agricultural technology adoption, ensuring content validity.

The four dimensions contain different numbers of indicators, ranging from four for Irrigation Technology to eight for Marketing Technology. To avoid disproportionate weighting caused by unequal item counts, indicator scores were first averaged within each dimension to produce standardized composite indices. These dimension-

level composite scores were then assigned equal weights in the PROMETHEE analysis. This two-stage aggregation approach ensures that each technological dimension contributes equally to the final ranking, regardless of the number of underlying indicators. Reliability analysis was conducted using Cronbach's alpha to assess the internal consistency of each technological dimension. The results indicate acceptable reliability for all constructs, exceeding the commonly accepted threshold of 0.70. Detailed reliability coefficients are presented in Appendix A.

4. Results and Discussion

Descriptive Statistics of Agricultural Technology Adoption

Table 1 presents the measurement structure of agricultural technology adoption across four key dimensions: TL, TI, TP, and TPr. Each dimension is operationalized through multiple indicators that capture specific aspects of technology use at the farm level.

Table 1. Measurement of agricultural technology adoption and descriptive statistics

Variable	Code	Indicator	Measurement Item	Mean	SD
Environmental Technology (TL)	TL1	Superior crop varieties	Use of high-yield crop varieties	2.68	1.399
	TL2	Fertilization practices	Application of site-specific fertilizers	3.20	1.446
	TL3	Balanced fertilization	Adoption of location-specific fertilization	2.38	1.282
	TL4	Soil testing	Conducting soil fertility tests	2.35	1.334
	TL5	Soil conservation	Implementation of soil conservation practices	2.32	1.373
Irrigation Technology (TI)	TI1	Drip irrigation	Use of drip irrigation systems	2.91	1.453
	TI2	Water harvesting	Use of water harvesting systems	2.58	1.515
	TI3	Pipe irrigation	Use of pipe-based irrigation networks	2.49	1.469
	TI4	Automated irrigation	Use of automated irrigation systems	2.74	1.390
Marketing Technology (TP)	TP1	Social media marketing	Use of social media for product sales	2.79	1.415
	TP2	Price monitoring	Monitoring market prices before selling	2.47	1.306
	TP3	Online marketplace	Selling via online platforms	2.18	1.293
	TP4	Digital finance	Access to digital financing platforms	2.11	1.431
	TP5	Online purchasing	Purchasing inputs via online platforms	2.66	1.368
	TP6	Online consultation	Consulting experts via digital platforms	2.44	1.340
	TP7	Intermediary sales	Selling through middlemen	2.26	1.338
	TP8	Computer use	Use of computers in farm management	2.51	0.908
Production Technology (TPr)	TPr1	Greenhouse	Use of greenhouse systems	2.44	1.005
	TPr2	Organic fertilizer	Production and use of organic fertilizer	2.61	0.977
	TPr3	Packaging innovation	Use of innovative packaging	2.49	0.826
	TPr4	Product processing	Processing agricultural products	3.56	1.096
	TPr5	ICM participation	Participation in ICM programs	2.39	0.742
	TPr6	IPM participation	Participation in IPM programs	2.41	1.664

Note: All measurement items were adapted from established literature and assessed using a five-point Likert scale. Reliability analysis indicates acceptable internal consistency across all constructs ($\alpha > 0.70$). Detailed reliability coefficients for each dimension are reported in Appendix A.

Overall, the mean values indicate a moderate level of technology adoption across all dimensions, with TPr showing relatively higher adoption, particularly in product processing (mean = 3.56). In contrast, TP exhibits comparatively lower adoption, especially in digital finance and online marketplace use, suggesting limited integration of digital tools into market-related activities. The standard deviation values, generally ranging from 0.8 to 1.6, indicate a moderate degree of variability among respondents, reflecting heterogeneous adoption patterns among farmers. This variation highlights differences in access to technology, capacity, and institutional support across regions. These findings provide an empirical foundation for subsequent multi-criteria analysis using the PROMETHEE method, as well as statistical validation and robustness testing, by demonstrating both the level and dispersion of agricultural technology adoption.

Socioeconomic Characteristics

The respondents' socioeconomic characteristics are categorized by age, education level, and farming experience. In terms of age, the distribution shows that 3% fall within the 15–20-year range, 21% within the 21–40-year range, 68% within the 40–60-year range, and 8% above 60 years.

Based on the socioeconomic characteristics shown in Figure 3, educational attainment varies: 5% did not attend school, 21% completed primary school, 46% completed junior high school, 24% completed senior high school, and 4% achieved a college education or higher. Most respondents therefore possess basic to intermediate formal education, a condition that may influence access to agricultural information, the development of management skills, and the adoption of technological innovations. Farming experience is also diverse. A total of 15% have less than 5 years of experience, 35% have 5–15 years of experience, 32% have 16–25 years of experience, and 18% have more than 25 years of experience in farming activities. The large proportion with longer experience reflects an

agricultural environment characterized by significant practical knowledge accumulation, long-term involvement, and reliance on traditional farming skills developed over time.

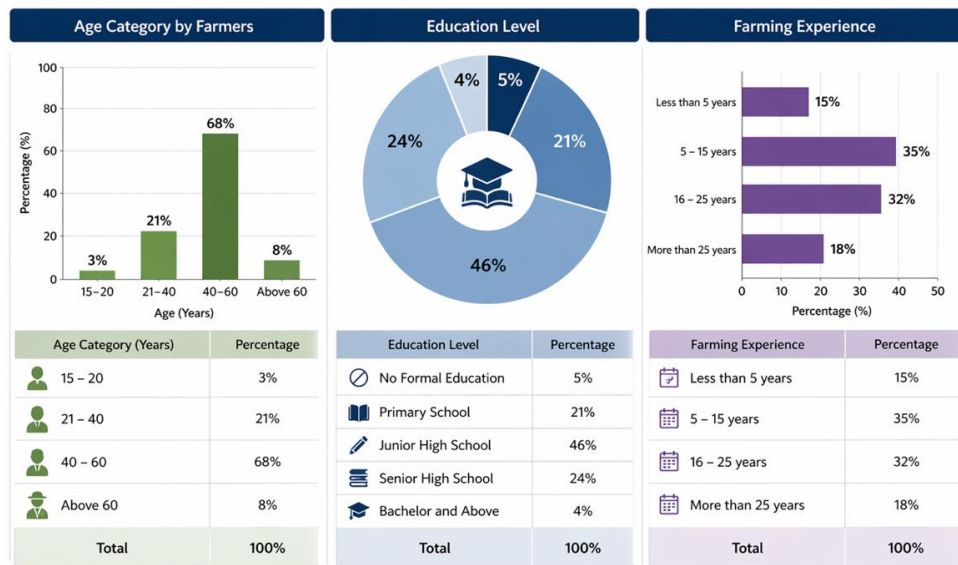


Figure 3. Socioeconomic characteristics of the respondents and their households, including age, formal education, and farming experience

4.1 Application of the Preference Ranking Organization Method for Enrichment Evaluation Method in Ranking Regional Agricultural Technology Competitiveness

The final ranking results and the network of alternative flows are presented in Figure 4 and Table 2. These scores represent the net preference flow derived from the PROMETHEE analysis, which integrates criteria weights, preference functions, and performance values. The results indicate that Batu City achieves the highest level of technological competitiveness, while Malang City and Malang Regency rank lowest. The increased weighting of productivity-related criteria positively influences the rankings of Batu City and Malang City, reflecting their strong performance in this dimension.

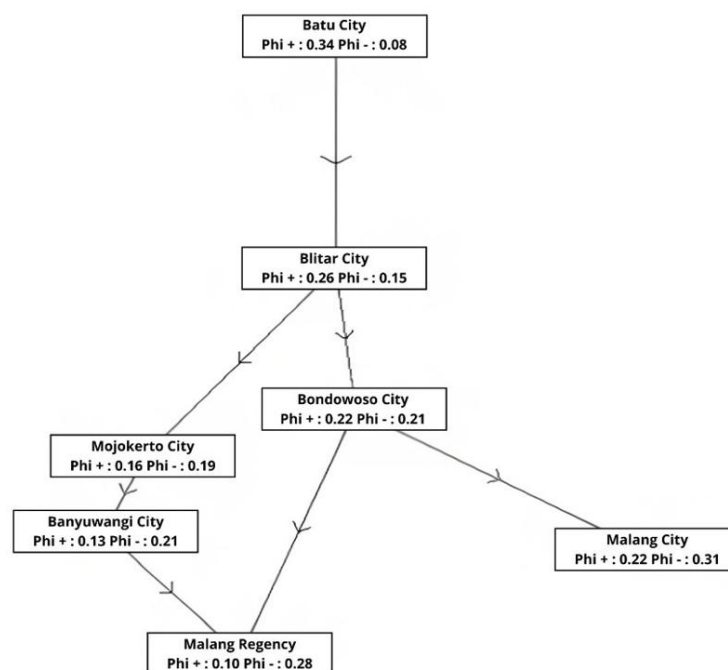


Figure 4. The Preference Ranking Organization Method for Enrichment Evaluation (POMETHEE) flow based on the region

Table 2. Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) II net flow scores, regional rankings, and rank stability analysis across technology dimensions

Region	PROMETHEE II Net Flow (Φ)				Average Net Flow (Φ)	PROMETHEE II Rank (Overall)	Rank Stability (Sensitivity Analysis 1–7)		Stability Index (1 = Most Stable)
	Environmental Technology	Irrigation Technology	Marketing Technology	Production Technology			Best Rank	Worst Rank	
Batu City	0.32	0.41	0.63	0.58	0.49	1	1	1	1.00
Malang City	0.21	0.34	0.49	0.37	0.35	2	2	2	1.00
Mojokerto City	0.07	0.31	0.39	0.22	0.25	3	2	3	0.83
Malang Regency	0.06	0.18	0.36	0.28	0.22	4	3	4	0.67
Blitar Regency	-0.18	0.05	0.23	0.07	0.04	5	4	6	0.50
Banyuwangi Regency	0.02	-0.06	0.12	-0.08	0.00	6	5	6	0.33
Bondowoso Regency	-0.29	-0.16	0.01	-0.22	-0.17	7	6	7	0.17

Technological competitiveness in agriculture varies substantially across regions in Indonesia, reflecting differences in technological integration and adoption. Batu City emerges as a leading region in East Java in relative technological competitiveness, as indicated by observed technology adoption indicators. This pattern may be associated with contextual factors discussed in previous studies, including greater access to modern technologies, higher digital readiness, and stronger institutional support. These factors collectively enhance the region's capacity to adopt and utilize advanced agricultural technologies.

The higher ranking of Batu City may reflect relatively stronger adoption across several technological dimensions represented in the survey indicators, particularly market-oriented and production-related technologies. Based on previous literature, such patterns may also be associated with broader technological integration, including digital platforms and modern farming practices. Such advancements contribute to increased productivity, improved resource efficiency, and stronger market linkages, thereby reinforcing overall regional competitiveness. The adoption of smart agriculture technologies has also been shown to enhance crop yields, reduce input costs, and mitigate climate-related risks, which are critical for long-term sustainability and resilience (Nguyen Thanh et al., 2021).

In contrast, Malang Regency demonstrates significantly lower technological competitiveness despite its geographical proximity to Batu City. The region is characterized by limited adoption of modern agricultural technologies, which is reflected in its negative net flow score. This disparity may reflect structural constraints discussed in the literature, including limited access to technology, lower adaptive capacity, and weaker institutional support for technology diffusion (Sudini & Wiryani, 2022). Efforts should focus on addressing structural barriers in less competitive regions by improving digital infrastructure, strengthening farmer capacity through training programs, and promoting incentives for the adoption of sustainable agricultural technologies. Reducing these disparities is essential not only for enhancing agricultural productivity but also for achieving broader development objectives, including food security, poverty reduction, and environmental sustainability (Bennett et al., 2013; Wicaksono, 2014).

Furthermore, competitiveness plays a critical role in regional development, particularly in the context of agricultural modernization and technological transformation (Mano Raj, 2021). Sensitivity analysis provides additional insights into the relative importance of different criteria, enabling a deeper understanding of how variations in weighting influence regional performance. By examining these dynamics, the analysis provides valuable insights into the key drivers of technological competitiveness, including technology adoption, environmental management, and innovation capacity (Emara et al., 2024).

Figure 5 presents a comparative PROMETHEE II analysis of agricultural technology adoption across selected regions. The radar charts illustrate the relative performance of each region across four technological dimensions: environmental, irrigation, marketing, and production. Regions with larger radar areas demonstrate stronger overall technological competitiveness and a more balanced adoption of multiple technologies. The accompanying table summarizes the net flow scores (Φ), regional rankings, and rank stability analysis. The results indicate that regions with higher competitiveness rankings tend to exhibit stronger adoption across several technological dimensions, particularly marketing and production technologies, whereas lower-ranked regions show weaker, less balanced technological performance.

Furthermore, the near-zero score of the environmental criterion in Batu City suggests that even highly competitive regions still face critical gaps in achieving sustainability. This indicates that technological advancement alone does not automatically translate into environmentally sustainable outcomes. To ensure long-term sustainability, leading regions must strengthen environmental management practices, particularly in resource

efficiency and ecological preservation (IPCC, 2022; Jones et al., 2019). This finding reinforces the argument that technological competitiveness should be complemented by environmentally balanced strategies to support sustainable agricultural development.

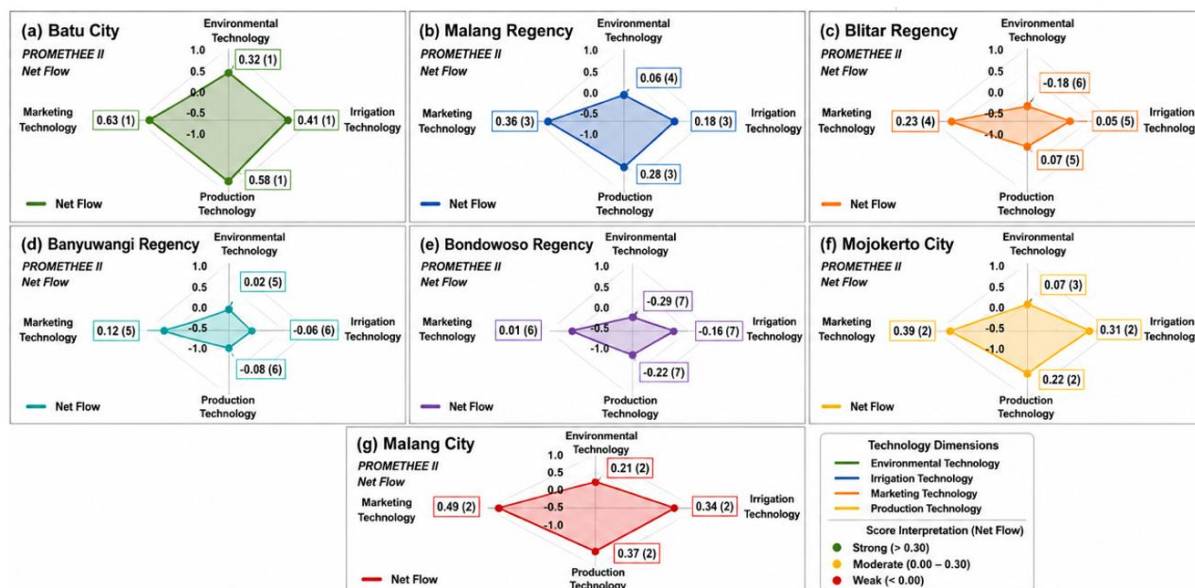


Figure 5. Comparative Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE II) net flow scores, ranking, and sensitivity stability of agricultural technology adoption across seven regions
Note: Net flow (Φ) ranges from -1 to +1. Higher values indicate better performance. Ranks are based on overall PROMETHEE II net flow under equal weights.

The empirical results further indicate that Batu City and Blitar Regency demonstrate relatively strong performance in adopting advanced agricultural technologies, particularly in environmental and irrigation dimensions. These regions have implemented practices such as site-specific nutrient management, soil fertility assessment, and the use of high-quality crop varieties adapted to local conditions, thereby improving efficiency and reducing environmental degradation. In addition, the adoption of automation technologies enhances operational efficiency in agricultural processes (Sujatha et al., 2023). Sustainable practices, including organic fertilizers, integrated crop–livestock systems, and crop diversification, further strengthen resilience and ecological sustainability (Gamage et al., 2023; Nguyen Thanh et al., 2021).

Irrigation technology also constitutes a key driver of competitiveness in these regions. The implementation of water harvesting systems, pipeline-based irrigation networks, and sensor-based automated irrigation contributes to more efficient water use and improved crop productivity. These findings are consistent with previous studies emphasizing the importance of irrigation efficiency and resource management in enhancing agricultural performance (Mwangi & Kariuki, 2015; Pham et al., 2021).

Moreover, Batu City and Blitar Regency have increasingly integrated digital technologies into marketing and financial systems. The utilization of online platforms enables farmers to access broader markets and reduce dependence on intermediaries, while digital financing mechanisms improve access to capital for technological upgrading (Feng et al., 2022; World Bank Group, 2019). In addition, digital advisory services enhance farmers' decision-making capacity and support more efficient farm management practices (Huang, 2023).

In contrast, regions such as Mojokerto City, Banyuwangi Regency, Malang City, and Malang Regency, which fall into the low-competitiveness category, exhibit relatively limited adoption of advanced technologies. Technological implementation in these areas tends to be fragmented and small-scale, resulting in lower impacts on productivity and sustainability. This disparity highlights structural challenges, including limited access to financial resources, weak institutional support, and inadequate knowledge transfer systems (Gao et al., 2020; Mponela et al., 2016). Consequently, targeted policy interventions are required to promote scalable and sustainable technology adoption, particularly in irrigation systems, environmental management, and digital agriculture.

Overall, the results reveal substantial regional disparities in agricultural technology competitiveness. Regions with higher rankings tend to perform more strongly across multiple dimensions, particularly in market-oriented and resource-efficient technologies. Conversely, lower-ranked regions exhibit weaker performance, reflecting constraints in technology access, institutional capacity, and market integration. These findings are consistent with the broader literature on agricultural transformation in developing countries (Barrett, 2021; Rahaman et al., 2021).

The methodological approach employed in this study enables a systematic comparison of regional

competitiveness across multiple technological criteria. The PROMETHEE method is widely recognized for its robustness in multi-criteria evaluation, allowing for the identification of both strengths and weaknesses across different dimensions (Behzadian et al., 2010; Brans & Vincke, 1985; Taherdoost, 2023). The findings underscore the critical role of agricultural technology integration in enhancing both competitiveness and sustainability (Bennett et al., 2013; Gao et al., 2020; Wang et al., 2023). Furthermore, the results highlight that competitiveness rankings are sensitive to the selection of criteria and weighting schemes, emphasizing the importance of carefully defining evaluation indicators in sustainability assessments (Cinelli et al., 2014; Hyde et al., 2003).

4.2 Inter-Regional Differences in Agricultural Technology Adoption: Evidence from Analysis of Variance

A one-way ANOVA was conducted to assess whether differences in technology adoption across regions are statistically significant, complementing the PROMETHEE analysis. The ANOVA results provide statistical support for the hypothesis that the underlying technology-adoption indicators differ across regions (Table 3). The results show that all variables exhibit significant differences ($p < 0.01$), indicating that regional variation in technological competitiveness is systematic rather than random. This finding complements the PROMETHEE ranking by demonstrating that significant inter-regional differences exist in the underlying technology-adoption indicators used in the multi-criteria analysis (Behzadian et al., 2010; Cinelli et al., 2014; Talukder & Hipel, 2018).

Table 3. Analysis of Variance (ANOVA) results for regional technology adoption

Variable	F-Value	p-Value	Interpretation
Environmental Technology (TL)	4.32	0.001	Significant differences across regions
Irrigation Technology (TI)	3.87	0.002	Significant differences across regions
Marketing Technology (TP)	5.11	0.001	Highly significant differences
Production Technology (TPr)	4.76	0.001	Significant differences across regions

Among the four dimensions, TP records the highest *F*-value, indicating that disparities in market access and commercialization capabilities are the most pronounced across regions. This finding is consistent with the literature highlighting the critical role of digital agriculture and market integration in shaping regional competitiveness (Aliev et al., 2023; World Bank Group, 2019). Overall, the ANOVA results confirm substantial inter-regional differences in technology adoption, thereby providing complementary statistical evidence for the comparative patterns identified through the PROMETHEE ranking.

Although statistically significant, these results do not imply causal relationships, possibly related to the cross-sectional nature of the data. Instead, they reflect structural differences in technology adoption, influenced by factors such as resource access, institutional support, and knowledge dissemination (Gao et al., 2020; Mponela et al., 2016; Mwangi & Kariuki, 2015). To further examine pairwise differences, a Tukey HSD post hoc test was conducted (Table 4).

Table 4. Post Hoc (Tukey HSD) comparison of selected regions

Comparison	Mean Difference	p-Value	Interpretation
Batu vs. Malang Regency	0.85	0.001	Significant
Batu vs. Mojokerto	0.62	0.002	Significant
Blitar vs. Malang Regency	0.54	0.005	Significant

The post hoc results confirm that Batu City significantly outperforms several regions, particularly Malang Regency and Mojokerto. The largest disparity is observed between Batu City and Malang Regency (mean difference = 0.85; $p = 0.000$), indicating a substantial competitiveness gap. These findings reinforce the presence of structural inequalities in technological adoption, particularly in the marketing and production dimensions, and are consistent with prior studies that emphasize the role of institutional capacity and innovation systems in shaping agricultural technology adoption (Gao et al., 2020; Mwangi & Kariuki, 2015; Wang et al., 2023).

4.3 Sensitivity and Robustness of Ranking Results under Alternative Weighting Scenarios

The sensitivity analysis was conducted to evaluate the robustness of the PROMETHEE ranking by varying the weights of four criteria: TL, TI, TP, and TPr (Table 5). The results demonstrate that regional rankings remain largely stable under $\pm 10\%$ and $\pm 20\%$ weight variations, indicating a high level of robustness in the composite index.

Batu City consistently ranks first across all scenarios, while Malang Regency remains the lowest-ranked region, confirming the stability of extreme positions. Blitar City and Bondowoso Regency also maintain their positions as second and third, respectively. Minor variations are observed only in the middle rankings, particularly between

Mojokerto and Malang City when the weight of TP is increased. This suggests that marketing-related criteria are relatively more sensitive in influencing ranking outcomes. In contrast, variations in environmental (TL), irrigation (TI), and production (TPr) weights do not significantly alter the ranking structure.

Table 5. Sensitivity analysis of regional rankings under varying weight scenarios

Scenario	TL	TI	TP	TPr	Batu City	Blitar City	Bondowoso	Mojokerto	Banyuwangi	Malang City	Malang Regency
Baseline (Equal Weights)	0.25	0.25	0.25	0.25	1	2	3	5	6	4	7
Scenario 1 (+10% TL)	0.30	0.23	0.23	0.24	1	2	3	5	6	4	7
Scenario 2 (+20% TL)	0.35	0.22	0.22	0.21	1	2	3	5	6	4	7
Scenario 3 (+10% TI)	0.23	0.30	0.23	0.24	1	2	3	5	6	4	7
Scenario 4 (+20% TI)	0.22	0.35	0.22	0.21	1	2	3	5	6	4	7
Scenario 5 (+10% TP)	0.23	0.23	0.30	0.24	1	2	3	4	6	5	7
Scenario 6 (+20% TP)	0.22	0.22	0.35	0.21	1	2	3	4	6	5	7
Scenario 7 (+10% TPr)	0.23	0.23	0.24	0.30	1	2	3	5	6	4	7
Scenario 8 (+20% TPr)	0.22	0.22	0.21	0.35	1	2	3	5	6	4	7

Notes: Environmental Technology (TL), Irrigation Technology (TI), Marketing Technology (TP), and Production Technology (TPr) Weights are proportionally adjusted (sum = 1). Rankings indicate performance (1 = highest).

To further assess stability, rank correlation analysis was conducted using Spearman's ρ and Kendall's τ (Table 6).

Table 6. Robustness test results of regional ranking

Scenario	Spearman's ρ	Kendall's τ	Robustness Level	Interpretation
Scenario 1 (+10% TL)	1.000	1.000	Perfect	No ranking change
Scenario 2 (+20% TL)	1.000	1.000	Perfect	Fully consistent
Scenario 3 (+10% TI)	1.000	1.000	Perfect	Fully stable
Scenario 4 (+20% TI)	1.000	1.000	Perfect	No structural shift
Scenario 5 (+10% TP)	0.964	0.905	Very Strong	Minor rank change
Scenario 6 (+20% TP)	0.964	0.905	Very Strong	Slight sensitivity
Scenario 7 (+10% TPr)	1.000	1.000	Perfect	Fully robust
Scenario 8 (+20% TPr)	1.000	1.000	Perfect	Stable rankings

The robustness test confirms that ranking results are highly stable across most scenarios, with perfect agreement observed in 6 of 8 cases. Slight deviations occur only when marketing weights are increased (S5–S6), but correlation values remain high, indicating very strong consistency. This suggests that moderate variations do not significantly influence the ranking structure in weighting assumptions. Overall, the findings indicate that the PROMETHEE ranking structure is relatively stable under moderate weighting variations, suggesting that the comparative competitiveness patterns are robust within the assumptions of the adopted multi-criteria framework. The limited sensitivity observed under marketing-weighted scenarios highlights the strategic importance of market access as a key driver of regional agricultural technology competitiveness. These results strengthen the model's validity and support its use as a decision-making tool for regional policy analysis.

4.4 Policy-Relevant and System-Oriented Implications for Enhancing Competitiveness

The empirical results obtained from the PROMETHEE analysis provide a robust quantitative ranking of regional agricultural technology competitiveness across environmental, irrigation, marketing, and production dimensions. To translate these empirical findings into actionable policy insights, Figure 6 introduces a conceptual interpretation through a feedback loop perspective. Rather than serving as an additional empirical model, this systems-based framework elucidates how static differences in technology adoption may translate into dynamic sustainability outcomes over time.

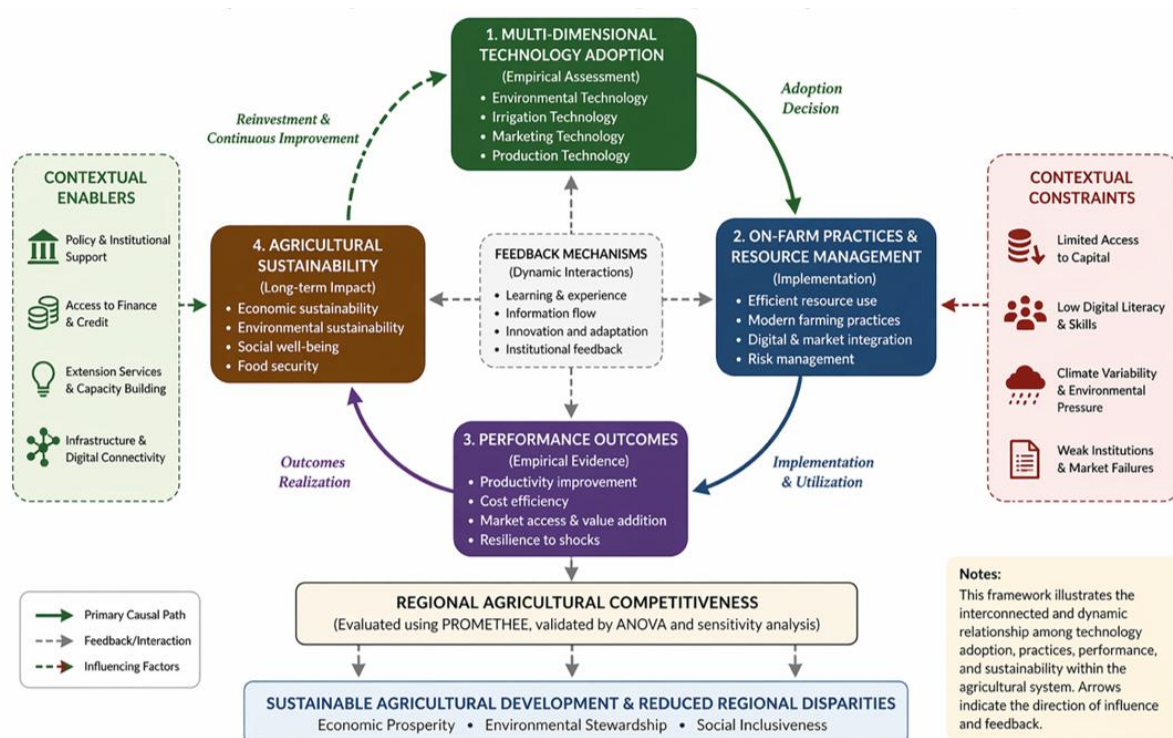


Figure 6. Conceptual framework: Feedback loop perspective on agricultural sustainability

Figure 6 presents a conceptual framework that interprets the empirical findings through a feedback loop perspective on agricultural sustainability. Rather than serving as an additional analytical model, this framework functions as an interpretive tool to explain the dynamic interactions among multidimensional technology adoption, on-farm practices, performance outcomes, and long-term sustainability. This perspective aligns with systems-based approaches emphasizing the interconnected and dynamic nature of agricultural systems and sustainability transitions (Bossel, 2002; Cinelli et al., 2014).

The framework is structured around four interconnected components. First, multidimensional technology adoption encompassing environmental, irrigation, marketing, and production technologies serves as the empirical foundation of the analysis. These dimensions, evaluated using the PROMETHEE method, reflect the extent to which regions integrate technological innovations into their agricultural systems. The application of multi-criteria decision-making approaches enables a comprehensive evaluation of complex sustainability and competitiveness structures (Behzadian et al., 2010; Taherdoost, 2023; Talukder & Hipel, 2018).

Second, on-farm practices and resource management capture the implementation stage, where adopted technologies are translated into operational activities. This includes efficient resource utilization, adoption of modern farming practices, digital integration, and risk management. The transition from adoption to implementation underscores that technological availability alone is insufficient; effective utilization can influence actual performance outcomes. This finding is consistent with studies emphasizing that technology adoption depends on knowledge transfer, farmer capacity, and institutional support (Gao et al., 2020; Mwangi & Kariuki, 2015; Wang et al., 2023). In addition, the growing role of digital agriculture further strengthens the link between technology and farm-level decision-making efficiency (Aliev et al., 2023; World Bank Group, 2019).

Third, performance outcomes conceptually represent possible implications of technology implementation, including potential improvements in productivity, cost efficiency, market access, and resilience to external shocks. However, these outcomes were not directly measured in the present study and are interpreted based on prior literature and contextual understanding. These outcomes are empirically reflected in the competitiveness rankings and validated through statistical analysis (ANOVA), reinforcing the significance of technological integration across regions. Previous studies suggest that integrated adoption of agricultural technologies may contribute to improvements in productivity, income, and resilience, particularly in developing country contexts (Amadu et al., 2020; Tesfaye et al., 2021; Tufa et al., 2019).

Fourth, agricultural sustainability is conceptualized as a long-term outcome encompassing economic viability, environmental integrity, and social well-being. The framework emphasizes that sustainability is not a direct result of individual technologies, but rather emerges from the cumulative and interactive effects of multiple technological dimensions. This multidimensional perspective aligns with the sustainability and food system transformation literature, which emphasizes the importance of integrated approaches to achieving long-term food security and

environmental balance (Braun & Ghosh, 2020; FAO, 2022; Rahaman et al., 2021).

A key contribution of this framework lies in its emphasis on feedback mechanisms that capture the dynamic, iterative nature of agricultural systems. Reinforcing loops (e.g., improved productivity leading to higher income and further technology adoption) illustrate how successful regions may experience cumulative advantages. Conversely, balancing loops highlights structural constraints such as limited access to finance, low digital literacy, and institutional weaknesses that may hinder adoption and slow system transformation. These dynamics are supported by studies on agricultural innovation and sustainability transitions, which emphasize path dependency and systemic interactions (Klerkx & Rose, 2020; Rotz et al., 2019).

In addition, the framework incorporates contextual enablers and constraints as external factors shaping the adoption–performance–sustainability pathway. Enablers, such as institutional support, financial access, extension services, and digital infrastructure, are discussed in the literature as factors that may facilitate technology uptake and strengthen agricultural system performance. In contrast, structural barriers may disrupt feedback processes and contribute to persistent regional disparities. Prior studies highlight that institutional quality, access to resources, and governance structures play a critical role in shaping agricultural transformation and regional competitiveness (Mponela et al., 2016; Pham et al., 2021; OECD/FAO, 2021).

Overall, this framework provides a theoretically grounded interpretation of the empirical results, demonstrating that regional agricultural competitiveness is a system-dependent phenomenon that may reflect the interaction of multiple technological, institutional, and socio-economic factors. By linking multi-criteria evaluation (PROMETHEE) with a systems-based perspective, the study offers a more comprehensive understanding of how technological adoption translates into sustainable agricultural development and reduced regional disparities. This integrative approach is particularly relevant in the context of digital transformation and climate change, where adaptive, coordinated strategies are essential to achieving resilient, sustainable agricultural systems (Feng et al., 2022; IPCC, 2022).

Figure 7 presents a systems-based and adaptive policy framework designed to enhance regional agricultural technology competitiveness and sustainability. The framework operationalizes the empirical findings derived from the PROMETHEE ranking, ANOVA testing, and sensitivity analysis into a structured set of policy interventions. It emphasizes that agricultural competitiveness is not a static outcome but a dynamic process shaped by continuous interactions between enabling conditions, technological adoption, implementation capacity, and adaptive governance mechanisms. The framework is organized into three interconnected policy domains: enabling policies, capacity-enhancing policies, and adaptive, feedback-driven policies. These domains reflect a sequential yet iterative process, consistent with the feedback loop perspective underlying the study.

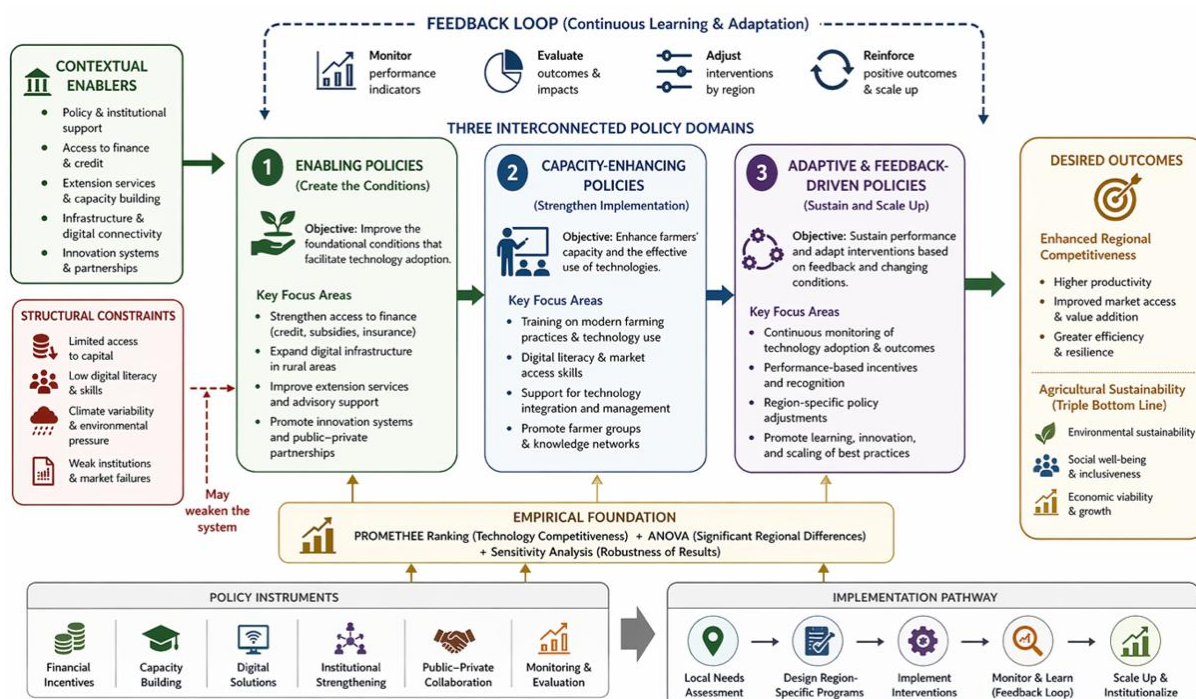


Figure 7. Policy framework for enhancing regional agricultural technology competitiveness and sustainability

First, enabling policies focus on establishing the foundational conditions necessary for technology adoption. These include access to finance, digital infrastructure, extension services, and institutional support. The empirical

findings indicate that regions with lower competitiveness tend to face structural barriers that hinder initial adoption. This is consistent with previous studies highlighting that access to resources and institutional quality are critical determinants of agricultural innovation uptake (Feder et al., 1985; Spielmann et al., 2011). Therefore, policy interventions at this stage should prioritize reducing entry barriers and improving accessibility to basic technological and financial resources.

Second, capacity-enhancing policies address the implementation gap between technology adoption and effective utilization. The results of this study show that higher-performing regions exhibit not only greater adoption but also more integrated use of technologies, particularly in marketing and production dimensions. This finding aligns with the concept of “technology bundling,” in which the combined use of complementary technologies yields higher productivity gains (Kilelu et al., 2013). Accordingly, policies should focus on strengthening farmers’ technical skills, promoting digital literacy, and supporting the integration of technologies across the value chain. Capacity-building initiatives, such as training programs and farmer networks, play a crucial role in translating technological potential into measurable performance outcomes.

Third, adaptive and feedback-driven policies emphasize the importance of continuous learning and policy adjustment. The feedback loop perspective suggests that improvements in productivity and income can reinforce further technology adoption, creating a virtuous cycle of competitiveness. However, this process is not automatic and may be disrupted by external shocks, such as climate variability or market instability. Therefore, adaptive governance mechanisms such as monitoring and evaluation systems, performance-based incentives, and region-specific policy adjustments are essential to sustain long-term competitiveness (Hall et al., 2001; World Bank Group, 2019). This approach is consistent with the broader literature on agricultural innovation systems, which highlights the need for iterative learning and institutional flexibility.

In addition to these policy domains, the framework incorporates contextual enablers and structural constraints as critical external factors influencing the effectiveness of interventions. Enablers, such as strong institutions, access to finance, and infrastructure, facilitate the smooth functioning of the adoption-implementation-performance cycle. In contrast, constraints such as limited capital, low digital literacy, and weak market systems can create negative feedback loops that trap regions in low-productivity equilibria. This dual perspective underscores the importance of addressing both opportunities and barriers in policy design.

From a broader perspective, the framework demonstrates that enhancing agricultural competitiveness requires a system-based policy approach that integrates multiple dimensions of technology and governance. Rather than focusing on isolated interventions, policymakers should adopt coordinated strategies that simultaneously address access, capacity, and adaptability. This finding supports the argument that sustainable agricultural development is inherently multidimensional, encompassing economic efficiency, environmental sustainability, and social inclusiveness (Rob et al., 2017; Pretty, 2008).

Furthermore, the policy framework contributes to the literature by linking multi-criteria decision analysis (PROMETHEE) with policy design and systems thinking. While PROMETHEE provides a robust tool for ranking regional competitiveness, the framework extends its application by translating quantitative results into actionable policy pathways. This integration enhances the study’s practical relevance and serves as a bridge between empirical analysis and policy implementation. Overall, the proposed framework offers a comprehensive and scalable model for improving regional agricultural competitiveness. By combining empirical evidence with a feedback-oriented policy perspective, it provides actionable insights for designing targeted, adaptive, and context-sensitive interventions to achieve sustainable agricultural development and reduce regional disparities.

5. Conclusion

This study provides empirical and methodological evidence that relative regional agricultural technological competitiveness, as measured through self-reported technology adoption indicators, is multidimensional and varies across regions. By applying an integrated framework combining PROMETHEE, ANOVA, and rank-robustness analysis, the study demonstrates that the comparative ranking structure remains relatively stable across alternative weighting assumptions, reinforcing the reliability of the findings. A key insight emerging from the analysis is that regions with higher relative competitiveness scores tend to demonstrate more integrated adoption across multiple technological dimensions, rather than systemic integration of these dimensions, particularly marketing and production capabilities. The sensitivity of rankings to marketing-related criteria underscores the pivotal role of market access and digital integration as leverage points for policy intervention.

Methodologically, this study contributes by moving beyond traditional multi-criteria approaches through the incorporation of robustness testing using rank correlation measures. This integration enhances the credibility and replicability of multi-criteria evaluations in agricultural and sustainability research. From a policy standpoint, the findings suggest that reducing regional disparities requires a system-oriented, targeted intervention strategy that focuses on strengthening institutional support, improving access to digital and irrigation technologies, and enhancing farmers’ adaptive capacity. While its cross-sectional design and non-probabilistic sampling limit the study, it provides a strong foundation for future research to incorporate longitudinal data and dynamic modeling.

Overall, this research offers a robust, scalable, and policy-relevant framework for evaluating and improving agricultural technology competitiveness in developing economies. Importantly, the PROMETHEE results should not be interpreted as direct measures of sustainability performance, productivity, or income outcomes. Rather, they represent comparative regional competitiveness based on composite technology-adoption indicators. Therefore, the policy implications derived from this study should be interpreted as indicative and context-dependent rather than causal conclusions.

Author Contributions

Conceptualization, B.H.; methodology, D.A.; software, E.S. and M.P.K.; validation, B.H. and N.A.M.R.; formal analysis, N.A.M.R.; investigation, B.H.; resources, B.H. and M.P.K.; data curation, E. S.; writing—original draft preparation, B.H., D.A., and N.A.M.R.; writing—review and editing, B.H. and D.A.; visualization, M.P.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Table A1. Reliability analysis of technological dimensions

Dimension	Number of Indicators	Cronbach's α
Environmental Technology (TL)	5	0.812
Irrigation Technology (TI)	4	0.784
Marketing Technology (TP)	8	0.861
Production Technology (TPr)	6	0.803

Note: All Cronbach's α coefficients exceed the minimum acceptable threshold of 0.70, indicating satisfactory internal consistency.